

NLP - assignment 3

1. Explain the Concept of word embedding in natural language processing (NLP). how does word embedding represent words in a continuous vector space.

— Word embedding is an advanced technique in natural language processing used to convert words into meaningful numerical representation. Unlike traditional method where words are treated as separate entities, word embedding capture the semantic meaning and relationships between words.

Advantages of word embeddings:

- Reduce dimensionality compared to one-hot encoding.
- Capture semantic meaning
- Improves performance in NLP tasks

popular models used for word embeddings include Word2vec, which uses neural networks and Glove, which uses global word co-occurrence statistics.

Applications:

- Sentiment analysis
- Machine translation
- Text classification
- Chat bot

Thus, word embedding represents words in a continuous vector space where distance and direction between vector reflect meaningful relationship between words.

Word embedding also capture semantic relationship through vector operation. for example: king - man + woman = queen. This show that embedding can understand analogies.

2. Describe the process of text vectorization in NLP. Explain how textual data is converted into numerical form and discuss the importance of vectorization in machine learning and language processing tasks.

- Text vectorization is the process of converting textual data into numerical format so that machines can understand and process it. Since computer cannot directly interpret text, this step is essential in all NLP-based machine learning models.

The process of text vectorization involves several steps:

1. Text processing: Before vectorization, the text is cleaned.
 - Removing punctuation and special characters.
 - Converting text to lowercase.
 - Removing stopwords (e.g.: "is", "the").
 - Stemming or lemmatization.

2. Tokenization: The cleaned text is split into smaller units called tokens (words or phrases).
Example: "I love NLP" → ["I", "love", "NLP"]

3. Vocabulary Creation

A set of unique words is created from the dataset.

4. Numerical Representation Techniques:

Different methods are used to convert tokens into numbers.

- Bag of words (BoW): Counts frequency of each word.
- TF-IDF: Weights words based on importance.
- Word embeddings: Dense vector representation.

5. Vector Formation:

Each document is converted into a numerical vector based on selected technique.

Machine learning models require numerical input.

- help in feature extraction from text.
- Enables comparison between documents.
- Improves model performance.

Application:

- Spam detection
- Sentiments analysis
- Document classification
- Recommendation system

Thus, text vectorization is a crucial step that transforms raw text into structured numerical data for effective machine learning processing.

What is Term Frequency (TF) in Natural language processing? Explain how TF is calculated and discuss its significance in identifying important words within a document.

- Term frequency (TF) is a Statistical measure used in Natural Language Processing to evaluate how important a word is within a specific document. It focuses only on the frequency of the word in that document.

The basic idea is that the more frequently a word appears in a document, the more important it is likely to be for that document.

Formula of TF:

TF = $\frac{\text{Number of times a word appears in a document}}{\text{Total number of word in the document}}$

Example: If a word appear 5 times in a document of 100 words.

$$TF = \frac{5}{100} = 0.05$$

Types of TF:

- Raw Count (Simple frequency)
- Normalized frequency
- Log-Scaled frequency

Significance of TF:

1. Identifies important words in a document.

2. Help in keyword extraction

3. used in search engines for ranking

4. Simple and Computationally efficient

Limitations: Does not consider important across multiple document.
Common words may get high TF unnecessarily

Application: Information retrieval
Text mining
Document ranking

Thus, TF is a foundational concept that helps measure word importance within a single document.

4. Define Inverse Document Frequency (IDF) in NLP.
Explain how describe its role in reducing the importance of common words across multiple documents.

Inverse Document Frequency (IDF) is a measure used in Natural language processing to evaluate how important a word is across a collection of documents (corpus). It helps reduce the influence of commonly used words.

While TF measure frequency within a document, IDF measure how unique or rare a word is across all documents.

Formula of IDF:

$$IDF = \log \left(\frac{\text{Total number of documents}}{\text{Number of documents containing the word}} \right)$$

Example: If a word appears in all document
IDF is low.
If it appears in very few document, IDF
is high.

Role of IDF:

- Reduces importance of common words like "the", "is", "and".
- Highlights rare and meaningful words.
- Improves text analysis accuracy.

Combined Measure (TF-IDF): TF & IDF are often Combined

$$TF-IDF = TF \times IDF$$

This ensure that: words frequent in a document but rare in corpus get high weight.

- Common words get low weight

Advantages: Improves search engine relevance
enhance classification models.
Reduce noise in text data

Applications: Search engine (Google, Bing)
Document Similarity
Text Classification

Thus, IDF plays a crucial role in balancing words importance across documents and improving the effectiveness of NLP models.